

Runtime Systems

Malloc, garbage collection. Liveness is undecidable, so collectors compute reachability instead.

MALLOC

Selfie's malloc is a bump pointer. Freeing memory makes everything harder.

```
// emit_malloc, in spirit
uint64_t* malloc(uint64_t size) {
    uint64_t* block = _bump; // read
    _bump = _bump + size; // modify, write
    if (_bump > brk) brk(_bump); // ask the kernel for pages
    return block;
}
```

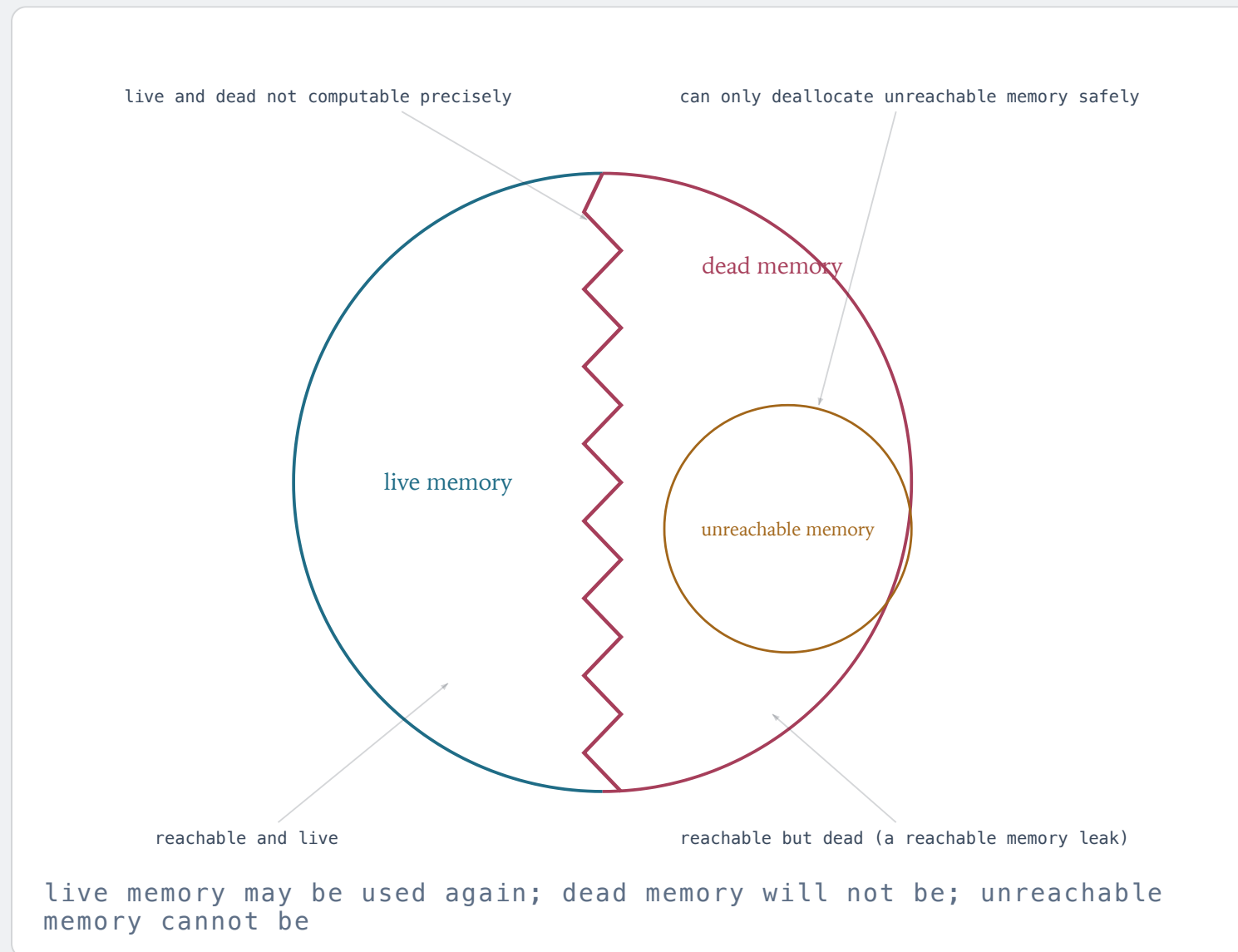
A global pointer that moves up. Allocation is constant time, memory is never reused, and the heap grows until the kernel refuses. Selfie's own compiler runs on this and never frees a byte; for a compiler that is fine.

Add free and you need a free list, a fit policy, coalescing against fragmentation, and metadata per block: the book's tour from first-fit to slab allocators. Every design is a different notation for the same meaning, a function from requests to disjoint blocks, with a different cost profile.

And the read-modify-write of `_bump` is last week's race. Two threads can receive the same block. That is the assignment.

Will this block be used again?

Undecidable. Can it be reached? Decidable.

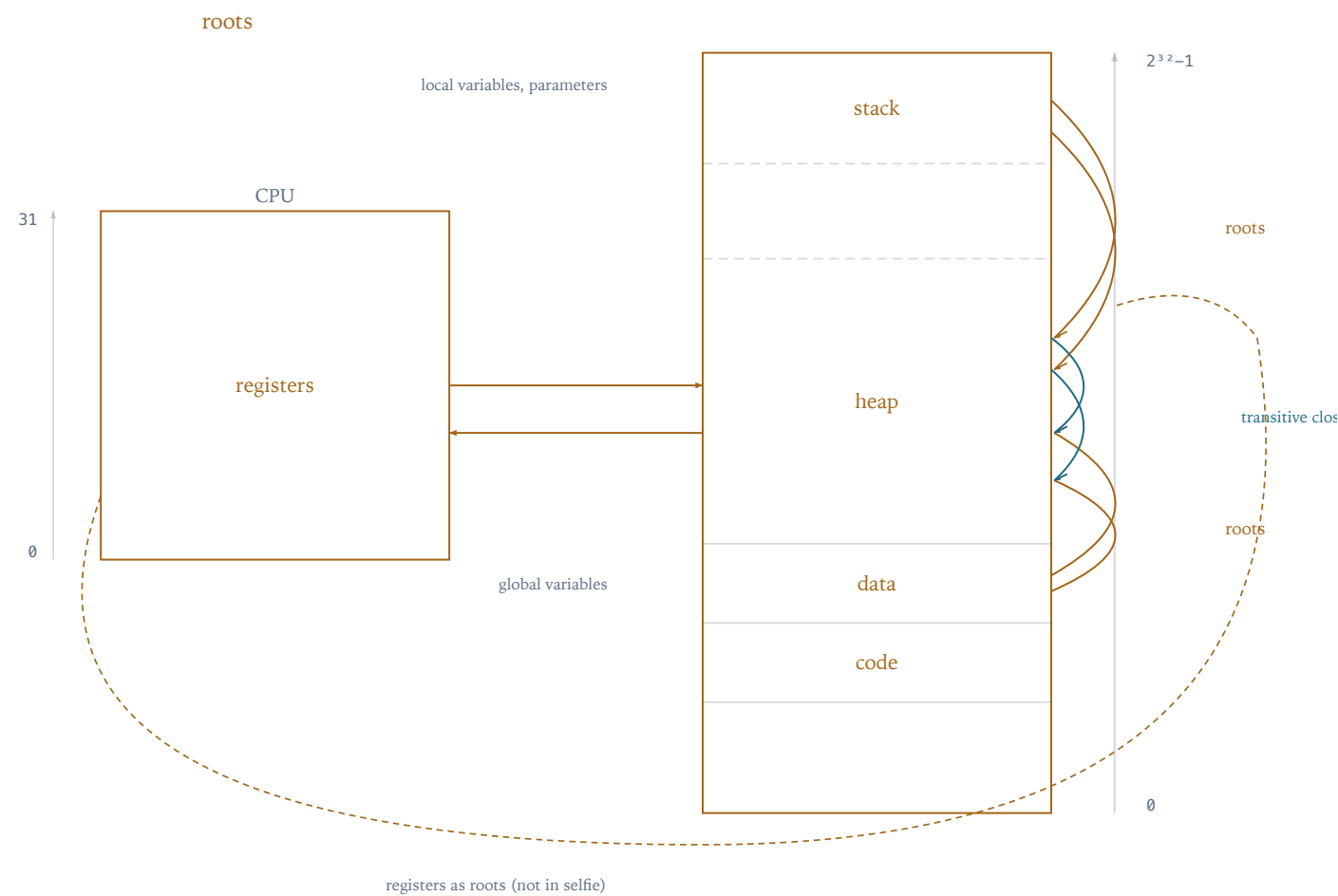


A block is live if the program will use it again and dead if it will not. Whether a program will ever use a given address is a property of its future behaviour: Rice, exactly. No collector decides liveness.

A block is reachable if some chain of pointers leads to it from a variable. Reachability is a property of the current state, computed by a graph traversal in time linear in the heap. Unreachable implies dead; the converse fails, and the difference is the memory a collector keeps although it is dead.

So every garbage collector computes the decidable over-approximation of the undecidable property. Safe, never precise. The design principle of the class again: bound what you cannot decide.

Start from the variables. Follow every word that looks like a pointer.



roots in the data and stack segments; the heap reached from them

The roots are the global variables in the data segment and the locals and parameters on every stack. Mark everything reachable from them, transitively; sweep everything unmarked. Mark-and-sweep, and selfie's collector does this on the emulator's view of the guest's memory, or as a library inside the program, or both at once.

Which words are pointers? C* does not say; a word is a number. A conservative collector treats every word that could be a heap address as if it were one: an over-approximation of an over-approximation, still safe, still a bound. Boehm's collector made that fast; selfie's makes it readable.

Self-collecting selfie: the emulator's collector collects the memory of a guest that is running its own collector on the memory of its guest. Try `-gc` at two levels and read the profile.

COST

Time, space, and the pause.

STOP-THE-WORLD

Mark-and-sweep stops the program for a time linear in the heap. Fine for a compiler; not for a game, a pacemaker, or a kernel. Every collector since is a way to spread that time out.

**GENERATIONAL,
INCREMENTAL, CONCURRENT**

Most objects die young: collect the young ones often. Do a little marking per allocation. Mark while the program runs, with a barrier that tells the collector what changed. Each is a bound traded for a cost.

REFERENCE COUNTING

Count pointers instead of tracing; free at zero. No pause, but cycles are never freed: an under-approximation of unreachability. Safe in the other direction: leaks, never dangling pointers.

Managed languages are languages whose runtime does this for you, and the price is paid on every allocation. The Cost chapter's trade, once more.

ASSIGNMENT

threadsafe-malloc: add `lr.d` and `sc.d`, then use them.

- 1 Add `lr.d` and `sc.d` to RISC-U: encoding and decoding in selfie, a case each in mipster's execute, and the reservation, cleared by any store to the address, in the machine context. Make hypster honour it too.
- 2 Provide them to C* as procedures `lr()` and `sc()`, emitted like the other library procedures.
- 3 Rewrite `emit_malloc` so that the read-modify-write of `_bump` is a retry loop on `lr` and `sc`. Test with last week's threads: many threads, many allocations, no two blocks overlapping.
- 4 `./grader/self.py threadsafe-malloc.`

NEXT WEEK

The last systems assignment, `treiber-stack`, builds a lock-free stack on the same two instructions. And `rotor-bounds` turns a model checker on your sequential systems code, because rotor does not yet model threads. Both are introduced in week 10.